

Competitive Pricing Analysis System

User Guide

Python-Based Competitive Pricing Analytics
and Market Positioning Evaluation Framework

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Core Technologies:

Python • Pandas • OpenPyXL

Development Environment:

PyCharm

Version Control & Repository:

Git • GitHub

Version:

1.0

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1. Project Overview

The Competitive Pricing Analysis System is a Python-based business analytics tool designed to evaluate a company's pricing position relative to multiple competitors.

The system processes company and competitor pricing datasets, compares prices across comparable products, and generates automated Excel reports containing pricing insights, charts, and strategic observations.

The analysis process is fully automated, requiring only the input pricing files and a user-defined pricing threshold.

The tool applies automated classification logic to identify:

- exclusive products;
- underpriced products;
- overpriced products;
- market-aligned products.

The generated reports include:

- pricing category classification;
- pricing distribution analysis;
- item category analysis;
- pricing group segmentation;
- pricing structure assessment;
- pricing risk evaluation;
- strategic recommendations;
- overall assortment positioning;
- automated Excel charts and tables.

The project combines data analysis, competitive pricing intelligence, automated reporting, and business decision-support functionality within a unified analytical workflow.

The system was developed to automate competitive pricing analysis, improve pricing transparency, and support data-driven business decision-making.

2. Required Input Files

The system requires a ZIP archive containing exactly 4 Excel (.xlsx) files:

- 1 company pricing file
- 3 competitor pricing files.

The names of both the ZIP archive and the Excel files may be arbitrary.

2.1 Company Pricing File

The company pricing file must contain exactly 3 columns:

Column Name	Description
item	Product/service name
item_category	Product category or product group
price	Product price without currency symbols

Example:

item	item_category	price
Product A	Category X	100
Product B	Category Y	350,11
Product C	Category X	130

Requirements:

- Each row must represent one product or service.
- Duplicate product names are not allowed within the same dataset.
- Prices must contain numeric values only: currency symbols, text values, and empty price cells are not supported.

2.2 Competitor Pricing Files

Each competitor pricing file must contain exactly 2 columns:

Column Name	Description
item	Competitor product/service corresponding to the company product
price	Competitor product price without currency symbols

Example:

item	price
Product A	99,99
Product C	140

Requirements:

- Product names in competitor files must either exactly match product names in the company pricing file or be omitted. Unmatched products will be excluded from the analysis.
- Not all products listed in the company pricing file are required to be present in competitor files.
- Prices must contain numeric values only: currency symbols, text values, and empty price cells are not supported.

3. How to Run the Program

1. Installation

The Competitive Pricing Analysis System can be run either with a local Python installation (Python 3.9 or later) or in a Python-compatible environment such as Google Colab, Visual Studio Code (VS Code), Jupyter Notebook, or PyCharm.

Before running the program, ensure that the required libraries are installed.
Open Terminal or Command Prompt and run:
`pip install pandas openpyxl`

Once the required libraries are installed, the system is ready for use.

2. Launch the Python program.

Run the main project file in the selected Python development environment.

3. Upload the ZIP archive.

When prompted, enter the path to the ZIP archive containing:

- 1 company pricing file,
- 3 competitor pricing files.

The system requires the path to a ZIP archive file. Providing a path to a folder or directory is not supported.

4. Select the company pricing file.

After the ZIP archive is processed, the system displays all detected Excel files.
Select the company pricing file by entering its corresponding number in the console.

5. Enter the pricing difference threshold.

When prompted, enter the pricing threshold percentage used for pricing category classification.

Decimal values must be entered using a dot (.) rather than a comma (,).
For example, enter 4.5 instead of 4,5.

The threshold determines how pricing differences between company and competitor products are evaluated.

6. Enter the output Excel filename.

Specify the filename for the generated pricing analysis report.

Examples:

- pricing_analysis_report.xlsx,
- final_pricing_report.xlsx

The system automatically generates the final Excel report.

4. Pricing Categories Logic

Before proceeding to the Generated Output section, users are strongly encouraged to review the Pricing Categories Logic presented below to ensure accurate interpretation of the generated analysis results.

The system classifies products into pricing categories based on the relationship between the company price and competitor prices.

Pricing comparison logic is evaluated using the user-defined pricing threshold percentage.

General pricing logic:

- “Significantly cheaper” means that the company price is lower than the competitor price by more than the defined threshold.
- “Significantly more expensive” means that the company price is higher than the competitor price by more than the defined threshold.
- “Slightly different” means that the pricing difference does not exceed the defined threshold percentage.

Pricing Category Definitions:

Category A is assigned when no competitor prices are available. Categories B–D are used when one competitor price is available. Categories E–K are used when multiple competitor prices are available.

Pricing Category	Definition
A	Exclusive product.
B	Significantly cheaper than one competitor.
C	Slightly different from or equal to one competitor.
D	Significantly more expensive than one competitor.
E	Significantly cheaper than multiple competitors.
F	Slightly different from or equal to the minimum competitor price.
G	Slightly different from both the minimum and maximum competitor prices.
H	Significantly higher than the minimum competitor price and significantly lower than the maximum competitor price.
I	Slightly different from or equal to the maximum competitor price.
K	Significantly more expensive than multiple competitors.

Pricing Category Group Definitions:

Pricing Category Group	Definition
A	Exclusive products.
B+E	Underpriced products.
D+K	Overpriced products.
C+F+G+H+I	Market-Aligned products.

5. Generated Output

Upon successful completion of the analysis, the system generates a multi-sheet Excel report containing detailed pricing analytics, visualisations, and strategic pricing insights.

The report consists of the following worksheets:

- Pricing Analysis Summary
- Pricing Analysis Data
- Item Category Analysis
- Filtered Pricing Categories (A, B+E, D+K, C+F+G+H+I)

Worksheet 1: Pricing Analysis Summary

The Pricing Analysis Summary worksheet provides a high-level overview of the pricing analysis results and serves as the executive summary of the report.

Pricing Category Group Distribution Chart

The worksheet includes a percentage-based bar chart showing the distribution of products across the four pricing category groups (A, B+E, D+K, C+F+G+H+I).

The chart enables rapid assessment of the overall pricing profile and dominant pricing patterns within the analysed assortment.

Pricing Analysis Summary Table

The worksheet also includes a summary table that provides key statistics and analytical findings derived from the pricing analysis.

The table contains:

1. Summary Statistics

Total products analysed & exact percentage values for each pricing category group displayed in the Pricing Category Group Distribution Chart.

2. Pricing Structure Assessment

The overall pricing structure identified by the system based on the distribution of products across pricing category groups.

3. Pricing Risk Assessment

Potential pricing risks associated with the identified pricing structure.

4. Overall Conclusion

An automated business interpretation summarising the company's pricing position, competitive alignment, and key strategic observations based on the analysis results.

Worksheet 2: Pricing Analysis Data

The Pricing Analysis Data worksheet contains the complete product-level dataset used throughout the pricing analysis process and serves as the foundation for all subsequent calculations, classifications, and report outputs.

Each row represents a single company product and includes:

- Product name (item)
- Product category (item_category)
- Company price (our_price)
- Competitor prices (competitor_price_1, competitor_price_2, competitor_price_3)
- Assigned pricing category (pricing_category)
- Pricing deviation (pricing_deviation)

Pricing Category

Each product is automatically assigned to one of the pricing categories (A–K) based on its pricing relationship to available competitor prices.

The assigned category reflects the product's competitive pricing position within the analysed market.

Pricing Deviation

The worksheet also reports the calculated pricing deviation between the company price and relevant competitor prices.

This metric helps quantify the magnitude and direction of pricing differences and supports the pricing classification process.

Missing Competitor Prices

Products that are not available in one or more competitor pricing files remain included in the analysis.

When a product listed in the company pricing file is not found in a competitor pricing file, the system automatically assigns a competitor price value of 0 for that competitor in the generated dataset. A competitor price value of 0 indicates that the product is unavailable from the corresponding competitor and is excluded from pricing comparisons.

The product remains included in the analysis and is evaluated using all available competitor price information.

Purpose

This worksheet serves as the primary analytical dataset for the entire report and provides complete transparency into the product-level calculations used to generate all pricing categories, charts, summaries, risk assessments, and strategic recommendations.

Worksheet 3: Item Category Analysis

The Item Category Analysis worksheet provides a detailed pricing distribution analysis for each product category (item_category) included in the dataset.

For every product category, the worksheet contains:

Pricing Category Distribution Analysis

A pricing category distribution analysis for the selected product category, consisting of a summary table and a corresponding distribution chart.

The analysis presents the percentage distribution of products across pricing categories A, B, C, D, E, F, G, H, I, and K.

The summary table includes:

- Pricing category;
- Percentage of products assigned to the category;
- Associated category colour group used in the chart visualisation.

The accompanying column chart visualises the same distribution data and displays:

- Pricing categories on the horizontal axis;
- Percentage of products on the vertical axis;
- Colour-coded pricing groups.

The analysis enables rapid identification of dominant pricing patterns within the selected product category.

Pricing Recommendations Table

A recommendation table summarising pricing behaviour at the product-category level.

The table contains the four pricing category groups (A, B+E, D+K, C+F+G+H+I).

For each pricing category group, the worksheet provides:

- Group interpretation;
- Signal strength assessment & percentage of products assigned to the group;
- Automatically generated pricing recommendations.

Signal strength is determined by the proportion of products assigned to each pricing category group and is classified as:

- Strong Signal (>25%)
- Moderate Signal ($\geq 10\%$ & $\leq 25\%$)
- Weak Signal (>0% & <10%)
- No Signal (0%)

The generated recommendations help identify potential pricing optimisation opportunities, evaluate pricing structure patterns within each product category, and support category-level pricing decision-making.

Worksheets 4-7: Filtered Pricing Category Worksheets

The worksheets A, B+E, D+K, and C+F+G+H+I provide filtered views of the Pricing Analysis Data, allowing users to review products grouped by broader pricing behaviour categories.

Each worksheet contains the same pricing-analysis fields as the Pricing Analysis Data but displays only products belonging to the corresponding pricing category group (A, B+E, D+K, C+F+G+H+I).

These worksheets are intended to simplify the identification and review of products exhibiting similar pricing behaviour and support more focused pricing analysis.

6. Error Handling

The system performs a series of validation checks before running the pricing analysis. If any validation rule is violated, the analysis process is stopped and an error message is displayed.

The following errors may occur:

Error Type	Description
ZIP File Not Found	The specified ZIP archive does not exist or the file path is incorrect.
Invalid ZIP Archive Content	The ZIP archive must contain exactly 4 Excel (.xlsx) files: 1 company pricing file and 3 competitor pricing files.
Invalid Company File Selection	An incorrect file number was entered when selecting the company pricing file.
Invalid Pricing Threshold	The pricing threshold must be a numeric value between 0% and 50%.
Missing Required Columns	One or more required columns are missing from the input files.
Invalid Table Structure	The input files do not follow the required dataset format described in this guide.
Non-Numeric Price Values	The price column contains text values, currency symbols, empty cells, or other non-numeric data.
File Access Error	The selected file cannot be accessed due to permission restrictions or because it is currently open in another application.
Invalid Output File Name	The output file name contains unsupported characters or cannot be created by the operating system.
Unexpected System Error	An unexpected error occurred during execution.

Validation Behaviour

If any validation error is detected:

- The analysis process is immediately terminated.
- No output report is generated.
- A descriptive error message is displayed to help identify and correct the issue.
- After correcting the problem, the analysis can be executed again.

7. Technologies Used

The Competitive Pricing Analysis System uses the following technologies and libraries:

Technology	Purpose
Python	Core programming language used to implement the pricing analysis system.
Pandas	Data loading, validation, transformation, and pricing analysis operations.
OpenPyXL	Excel workbook generation, worksheet formatting, and chart creation.
ZipFile	Extraction and processing of input Excel files from ZIP archives.
OS	File and directory management operations.
Shutil	Temporary file and folder management during data preparation.
Git - GitHub	Version control and source code repository.

Output Formats: The system generates analysis results in Microsoft Excel (.xlsx) format.

8. Source Code Repository

The complete source code for the Competitive Pricing Analysis System is available in the GitHub repository below.

GitHub Repository: <https://github.com/yooo011/competitive-pricing-analysis-system.git>

9. Future Improvements

A planned future enhancement of the Competitive Pricing Analysis System is the integration of sales volume analysis alongside the existing pricing analysis framework.

The current version evaluates products exclusively from a pricing perspective by comparing company prices with competitor prices. While this approach provides valuable insights into competitive positioning, it does not consider product sales performance.

Future versions of the tool may incorporate sales volume data to enable a more comprehensive evaluation of pricing effectiveness. By combining pricing position and sales volume metrics, the system would be able to identify products that are:

- competitively priced and generating strong sales;
- competitively priced but underperforming in sales;
- overpriced yet maintaining strong sales performance;
- underpriced without corresponding sales benefits;
- requiring pricing optimisation or strategic repositioning;
- and many other pricing and sales performance patterns.

The integration of sales volume analysis would allow the tool to move beyond pure price comparison and support more informed pricing decisions based on both competitive positioning and market performance.

This enhancement would provide a more holistic view of product performance and improve the quality of pricing recommendations, strategic assessments, and business decision-making.